

Direct English-to-Yoruba speech Translation model using Transformer with Augmented Attention Mechanism.

Olaniyan Julius
Department of Computer Science
Bowen University
Osun State, Nigeria
julius.olaniyan@bowen.edu.ng

Ibidun. C. Obagbuwa
Department of Computer Science &
Information Technology
Sol Plaatje University
Kimberley, South Africa
ibidun.obagbuwa@spu.ac.za

Esiefarienrhe B. Michael
Department of Computer Science
North-West University
Mafikeng, South Africa
25840525@nwu.ac.za

Ayodele A. Adebisi
Department of Computer Science
Landmark University
Omu-Aran,
Kwara State, Nigeria
ayo.adebiyi@lmu.edu.ng

Olaniyan Deborah
Department of Computer Science
Bowen University
Osun State, Nigeria
deborah.olaniyan@bowen.edu.ng

Sadura Priscilla Akinrinwa
Department of Data Science
Federal University of Technology
Akure
Akure, Nigeria
spakinrinwa@futa.edu.ng

Abstract—This study explores English-to-Yoruba speech-to-speech translation using a transformer neural network with 36 encoder-decoder pairs and 42 multi-head attention mechanisms. Unlike traditional cascaded models, this approach aligns speech directly, reducing dependency on large spectrogram datasets. Trained on a curated dataset of 200,000+ data points, the model achieved a BLEU score of 0.7, an MOS score of 2.8, and 98% translation accuracy, demonstrating its potential for practical speech translation applications.

Keywords—Speech-to-speech translation, Transformer neural network, Cascaded model, End-to-end model, BLEU score, MOS score

I. INTRODUCTION

Speech translation systems play a vital role in enabling cross-linguistic communication. While significant progress has been made for high-resource languages like English, French, and Spanish, low-resource languages such as Yoruba remain underexplored [1], [2]. Key challenges include limited parallel data, dialectal variations, and structural differences [3], [4]. Given that Yoruba is spoken by over 40 million people in West Africa, there is a strong need for effective English-to-Yoruba translation models [5], [6].

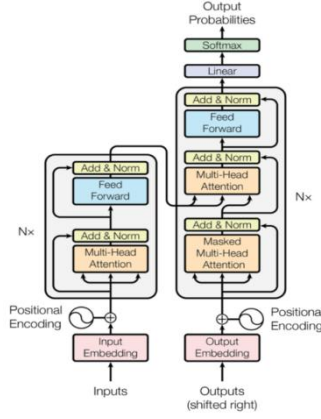
Recent advancements in deep learning, particularly the Transformer Neural Network (TNN), have greatly improved natural language processing tasks, including machine translation [8], [9]. The TNN model excels at capturing long-range dependencies in both text and speech, making it a promising model for speech translation [10], [11]. However, traditional Transformers struggle with the linguistic complexity of Yoruba, especially when trained on limited data [12]. To address this, a direct speech translation model is proposed, incorporating an augmented attention mechanism to better capture contextual dependencies and language-specific variations. This enhanced Transformer model integrates improved attention mechanisms and data augmentation techniques to overcome data scarcity and improve translation accuracy. This paper provides an in-depth analysis of the model's architecture, performance comparison with baseline models, and its potential impact on speech translation for low-resource languages. Several studies have explored direct Speech-to-Speech Translation (S2ST) using machine learning techniques. In 2020, [13] investigated an End-to-End S2ST system inspired by Google Translatotron, employing a sequence-to-sequence (STS) Encoder-Decoder model with spectrogram inputs. While the approach showed promise, it exhibited inconsistencies

compared to traditional cascaded models. In 2021, [14] proposed a direct S2ST system utilizing discrete units to improve efficiency and eliminate textual intermediaries. The model demonstrated superior performance on Spanish-English datasets, though its evaluation was limited to this language pair. Similarly, [15] introduced a Transformer-based S2ST model incorporating a transcoder to enhance frame-to-frame alignments. Despite improved translation for both syntactically similar and distant languages, the system still relied on spectrogram processing. Beyond speech translation, [16] developed a multi-lingual speech-to-Indian Sign Language (ISL) translation system, SISLA, employing a cascaded approach. This system aimed to bridge communication gaps for the hearing-impaired community by integrating speech recognition, machine translation, and ISL gesture synthesis.

Additionally, [17] explored an End-to-End speech translation approach using pre-trained models, leveraging Wav2Vec 2.0 as the encoder and mBART as the decoder. Their system performed well in translating English TED talk audio into written German text. A review of these studies highlights key challenges, including slow inference, limited speech datasets, and reliance on spectrogram-based training, which this research seeks to address.

II. METHODOLOGY

This study utilizes a Transformer model with an augmented attention mechanism for English-to-Yoruba speech-to-speech translation. The encoder processes English audio input, while the decoder generates Yoruba audio output. By leveraging self-attention, the model captures long-range dependencies efficiently, surpassing traditional RNNs through parallel processing. The encoder employs self-attention and feed-forward layers to learn complex word dependencies, while the decoder integrates an encoder-decoder attention mechanism to focus on relevant English representations. This architecture enhances the model's ability to learn intricate speech relationships, improving translation quality.



A. Existing Attention Mechanism

Self-attention plays a crucial role in the Transformer model as it allows capturing the dependencies between words or tokens in an input sequence. By computing attention weights through the dot product between a query vector and a set of key vectors, the model can assign importance to different parts of the sequence. In equation 10, the computed attention weights are scaled by the square root of the dimension of the key vectors. This scaling ensures that the attention values are appropriately adjusted to facilitate effective information flow within the self-attention mechanism.

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where:

A = Attention

Q = a matrix containing the Query

K = all Keys, the vector representations of all the words in the sequence.

V = the Value, which is often the same as K .

The V values are multiplied and summed with attention-weights a . The weights are defined using the following equation:

$$a = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (2)$$

The Softmax function normalizes the attention scores within a range of 0 to 1, which are then applied to the value matrix VVV . A standard transformer model incorporates three multi-head attention mechanisms: one in the encoder and two in the decoder, working together to optimize translation performance. As illustrated in Fig. 2, each input token in the encoder undergoes projection through trainable Query, Key, and Value weight matrices. This section provides a detailed explanation of the attention mechanisms in both existing and proposed transformer models, along with their mathematical formulations.

The existing projections for the encoder's attention mechanism are illustrated in equations (3), (4), and (5) respectively.

$$\begin{aligned} Q &= WQX_s(3) \\ K &= WKX_s(4) \\ V &= WVX_s \end{aligned} \quad (5)$$

where:

Q, K , and V are query, key and value matrices

W = projection weight

X_s = input token

Then, the attention score, a_s is computed as:

$$a_s = \frac{K^T * Q}{\sqrt{d_k}} \quad (6)$$

where:

d_k = the dimension of the key matrix.

The attention score (a_s) is then softmaxed to get the attention weights, a_w

$$a_w = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) \quad (7)$$

So, the output vector of one attention head is computed as:

$$eAM(Q, K, V) = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) * V \quad (8)$$

Similarly, the multi-head attention can be computed as follows:

$$eh_1 = eAM_1(Q_1 W_1^{Q_1}, K_1 W_1^{K_1}, V_1 W_1^{V_1}) \quad (9)$$

$$eh_2 = eAM_2(Q_2 W_2^{Q_2}, K_2 W_2^{K_2}, V_2 W_2^{V_2}) \quad (10)$$

$$eh_n = eAM_n(Q_n W_n^{Q_n}, K_n W_n^{K_n}, V_n W_n^{V_n}) \quad (11)$$

$$eMA(Q, K, V) = \text{concat}(eh_1, eh_2, \dots, eh_n)W \quad (12)$$

where:

eAM denotes one attention head output, $\{eh_1, eh_2, \dots, eh_n\}$ denote encoder attention heads ranging from first head to the n th head, eMA denotes encoder Multi-head Attention.

Similarly, the existing projections for the decoder's attention mechanism are illustrated in equations (13), (14), and (15) respectively.

$$Q = WQX_t(13)$$

$$K = WKX_t(14)$$

$$V = WVX_t \quad (15)$$

where:

Q, K , and V are query, key and value matrices for the target tokens, W is the projection weight, and X_t is the target token.

Then, the attention score, a_t is computed as:

$$a_t = \frac{K^T * Q}{\sqrt{d_k}} \quad (16)$$

where:

d_k is the dimension of the key matrix of the target token.

The attention score (a_t) is then masked to pre-empt the decoder from cheating. The masked attention, ma is computed using equation (17).

$$ma = \begin{bmatrix} a_{11} & -\text{inf} & -\text{inf} & -\text{inf} \\ a_{21} & a_{22} & -\text{inf} & -\text{inf} \\ a_{31} & a_{32} & a_{33} & -\text{inf} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \quad (17)$$

to normalize the masked attention score, the softmax function was used as in equation (18).

$$NW = \text{softmax} \left[\begin{bmatrix} a_{11} & -\text{inf} & -\text{inf} & -\text{inf} \\ a_{21} & a_{22} & -\text{inf} & -\text{inf} \\ a_{31} & a_{32} & a_{33} & -\text{inf} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \right] \quad (18)$$

$$NW = \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \quad (19)$$

where:

NW denotes normalized masked attention score.

So, to compute the self-attention of the first attention mechanism of the decoder, equation (20) is used.

$$dAM_1(Q, K, V) = NW * V \quad (20)$$

Each head is computed and then concatenated using the following equations:

$$dh_{11} = dAM_{11}(NW_1 * V_1) \quad (21)$$

$$dh_{12} = dAM_{12}(NW_2 * V_2) \quad (23)$$

$$dh_{1n} = dAM_{1n}(NW_n * V_n) \quad (24)$$

$$dMA_1(O, K, V) = \text{concat}(dh_{11}, dh_{12}, \dots, dh_{1n}) \quad (25)$$

where:

$dh_{11}, dh_{12}, \dots, dh_{1n}$ denote decoder attention heads ranging from first head to the n th head and dMA_1 denotes the first Multi-head Attention of the decoder.

For the second attention mechanism of the decoder, the output vector of the encoder is here projected into query and key matrices, while the output representation of the first attention mechanism is projected as the value matrix. So, both the query, key and value matrices are the inputs to the second attention mechanism of the decoder. This illustrated in equations (26), (27), and (28) respectively.

$$Q = eMAQ \quad (29)$$

$$K = eMAK \quad (30)$$

$$V = dMA_1V \quad (31)$$

With the above projections, it is easy to compute both the attention scores, a_s and weights, a_w respectively as illustrated in equations (32) and (33).

$$a_s = \frac{K^T * Q}{\sqrt{d_k}} \quad (34)$$

$$a_w = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) \quad (35)$$

By multiplying the attention weights and the value matrices, the first self-attention of the second attention mechanism of the decoder is computed as illustrated in equation (d17).

$$dAM_2(Q, K, V) = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) * V \quad (36)$$

Using equation (36), each attention head can be computed and later concatenated as illustrated in equations (37), (38), (39) and (40) respectively depending on the dimensionality of the attention head.

$$dh_{21} = dAM_{21}(a_{w1} * V_1) \quad (37)$$

$$dh_{22} = dAM_{22}(a_{w2} * V_2) \quad (38)$$

$$dh_{2n} = dAM_{2n}(a_{wn} * V_n) \quad (39)$$

$$dMA_2(Q, K, V) = \text{concat}(dh_{21}, dh_{22}, \dots, dh_{2n}) \quad (40)$$

where:

$dh_{21}, dh_{22}, \dots, dh_{2n}$ denote decoder attention heads ranging from first head to the n th head and dMA_2 denotes the second Multi-head Attention of the decoder. The multi-head attention mechanism is illustrated in Fig. 2.

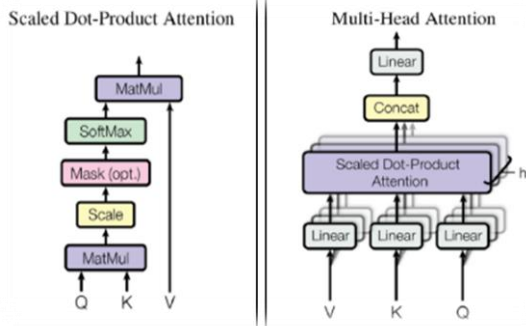


Fig. 2. Multi-Head Attention
(Source: Vaswani et al., 2017)

B. The Proposed Attention Mechanism

Similar to the existing attention mechanism of the transformer architecture, the proposed self-attention mechanism in this study also incorporated the technique of projecting the input vector into three learnable projection matrices namely; Query, Key, and Value matrices. The difference is that the value matrix is re-projected into two different other matrices in order to capture more input representation as depicted in Fig. 3.

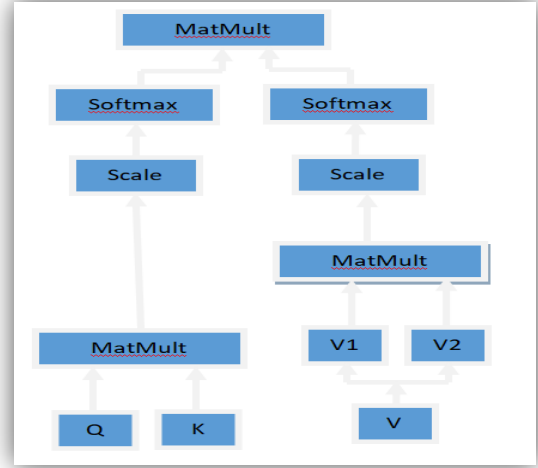


Fig. 3. Proposed Attention Mechanism

The enhanced value matrix is calculated to capture improved and richer representations compared to the original value matrix. This is achieved by transposing the second value matrix, multiplying it with the first value matrix, scaling the result to ensure stability during backpropagation, and then multiplying it by the initial value projection matrix. The resulting matrix is further multiplied by the computed attention weight matrix to generate a more refined and contextualized abstract representation of the input vector.

The proposed projections for the encoder's attention mechanism are illustrated in equations (41), (42), (43), (44) and (45) respectively. The new derived matrix, $newV$, is computed similar to the attention weights matrix computation as illustrated in equation (46). The proposed encoder attention mechanism for a single head is then computed as illustrated in equation (49).

$$Q = WQX_s \quad (41)$$

$$K = WKX_s \quad (42)$$

$$V = WVX_s \quad (43)$$

$$V_1 = WV_1V \quad (44)$$

$$V_2 = WV_2V \quad (45)$$

$$newV = \text{softmax}\left(\frac{V_2^T * V_1}{\sqrt{d_v}}\right) * V \quad (46)$$

$$a_s = \frac{K^T * Q}{\sqrt{d_k}} \quad (47)$$

$$a_w = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) \quad (48)$$

$$eAM(Q, K, V) = \text{softmax}\left(\frac{K^T * Q}{\sqrt{d_k}}\right) * \text{softmax}\left(\frac{V_2^T * V_1}{\sqrt{d_v}}\right) * V \quad (49)$$

$$eAM(Q, K, V) = a_w * newV \quad (50)$$

To compute for multi-heads, each head is computed and the concatenated as illustrated in equations (51), (52), (53) and (54) respectively.

$$eh_1 = eAM_1(a_{w1} * newV_1) \quad (51)$$

$$eh_2 = eAM_2(a_{w2} * newV_2) \quad (52)$$

$$eh_n = eAM_n(a_{wn} * newV_n) \quad (53)$$

$$eMA(Q, K, V) = \text{concat}(eh_1, eh_2, \dots, eh_n) \quad (54)$$

where:

$eh_1, eh_2, \dots, eh_n$ denote each attention head with eh_1 being the first and eh_n being the n th attention head.

eMA denotes the proposed encoder multi-head attention.

Similar to the encoder computation, the decoder is computed using the same projection but with different inputs, target embeddings. Here, the new value matrix is denoted as $dnewV$ as illustrated in equation (60).

$$Q = WQX_t \quad (55)$$

$$K = WKX_t \quad (56)$$

$$V = WVX_t \quad (57)$$

$$V_1 = WV_1V \quad (58)$$

$$V_2 = WV_2V \quad (59)$$

$$dnewV = softmax\left(\frac{V_2^T * V_1}{\sqrt{d_v}}\right) * V \quad (60)$$

where d_v is the dimension of the original value matrix which has the same dimensionality as V_1 and V_2 .

Then the attention score is computed as z where:

$$z = \frac{K^T * Q}{\sqrt{d_k}} \quad (61)$$

The attention scored is then masked by multiplying it with negative infinity, which will be zeroed out using softmax activation function as illustrated in equations (62) and (63) respectively.

$$mz = \begin{bmatrix} z_{11} & -inf & -inf & -inf \\ z_{21} & z_{22} & -inf & -inf \\ z_{31} & z_{32} & z_{33} & -inf \\ z_{41} & z_{42} & z_{43} & z_{44} \end{bmatrix} \quad (62)$$

where:

mz represents masked attention scores

$$nz = softmax\left[\begin{bmatrix} z_{11} & -inf & -inf & -inf \\ z_{21} & z_{22} & -inf & -inf \\ z_{31} & z_{32} & z_{33} & -inf \\ z_{41} & z_{42} & z_{43} & z_{44} \end{bmatrix}\right] \quad (63)$$

$$nz = \begin{bmatrix} z_{11} & 0 & 0 & 0 \\ z_{21} & z_{22} & 0 & 0 \\ z_{31} & z_{32} & z_{33} & 0 \\ z_{41} & z_{42} & z_{43} & z_{44} \end{bmatrix} \quad (64)$$

where:

nz denotes normalized masked attention scores.

Since nz and $dnewV$ are both normalized, then taking their product as illustrated in equation (65) will result in computing a single self-attention of the first decoder's attention mechanism, dA_1 .

$$dA_1(Q, K, V) = nz * dnewV \quad (65)$$

To compute for each head in a multi-head scenario, equations (66), (67), (68) were used and concatenated using equation (69) respectively.

$$dh_1 = dA_1M_1(nz_1 * dnewV_1) \quad (66)$$

$$dh_2 = dA_1M_2(nz_2 * dnewV_2) \quad (67)$$

$$dh_n = dA_1M_n(nz_n * dnewV_n) \quad (68)$$

$$dMA_1(Q, K, V) = concat(dh_1, dh_2, \dots, dh_n) \quad (69)$$

where:

$dh_1, dh_2, \dots, dh_n$ denote each attention head with dh_1 being the first and dh_n being the n th attention head of the first attention mechanism of the decoder.

dMA_1 denotes the proposed first multihead attention mechanism of the decoder.

Similar to the existing model, for the proposed second attention mechanism of the decoder, the output vector of the encoder is also projected into query and key matrices, while the output representation of the first attention mechanism is projected as the value matrix. So, both the query, key and value matrices are the inputs to the second attention mechanism of the decoder. This is illustrated in equations (70), (71), and (72) respectively.

$$Q = eMAQ \quad (70)$$

$$K = eMAK \quad (71)$$

$$V = dMA_1V \quad (72)$$

With those three projections, the attention weight, a_w is computed using equation (73).

$$a_w = softmax\left(\frac{K^T * Q}{\sqrt{d_k}}\right) \quad (73)$$

The resulting representation for a single head of the second attention mechanism of the decoder can then be computed as illustrated in equation (74).

$$dAM_2(Q, K, V) = a_w * V \quad (74)$$

To compute for each head in a multiheadmodel, equations (75), (76), (77) are used and concatenated using equation (p37) respectively.

$$dh_{21} = dA_2M_1(a_{w1} * V_1) \quad (75)$$

$$dh_{22} = dA_2M_2(a_{w2} * V_2) \quad (76)$$

$$dh_{2n} = dA_2M_n(a_{wn} * V_n) \quad (77)$$

$$dMA_2(Q, K, V) = concat(dh_{21}, dh_{22}, \dots, dh_{2n}) \quad (78)$$

where:

$dh_{21}, dh_{22}, \dots, dh_{2n}$ denote each attention head with dh_{21} being the first and dh_{2n} being the n th attention head of the second attention mechanism of the decoder.

dMA_2 denotes the proposed second multi-head attention mechanism of the decoder.

The Transformer neural network model was employed with carefully selected hyperparameters to optimize performance for speech translation. The input and output dimensionality were both set to 200, reflecting the spectrogram-based input features and classification output. To effectively capture complex patterns, the model used a hidden dimension of 512 and 36 transformer layers in both the encoder and decoder. To enhance attention mechanisms, the model utilized 42 attention heads, enabling efficient parallel attention operations. The feed-forward network (FFN) dimensionality was set to 2048, ensuring robust non-linear transformations. A dropout probability of 0.1 (10%) was applied to prevent overfitting. The model was trained using the Adafactor optimizer with a learning rate of 0.001 and a batch size of 64 for 300 epochs, employing mean square error (MSE) as the loss function. The dataset, comprising 200,023 sentences, was split in a 7:2:1 ratio, with 70% for training, 20% for testing, and 10% for validation. The model's performance was evaluated based on BLEU and MOS scores, confirming its effectiveness for the task.

C. Dataset

This research utilized a carefully curated dataset comprising English and Yoruba speech samples to train and evaluate a Transformer-based speech-to-speech translation model. The dataset was sourced from publicly available speech corpora and recorded native speaker samples, ensuring diversity across demographics, age groups, and regional variations. Ethical considerations were strictly followed, with informed consent obtained from all participants. The dataset included 200,023 audio recordings, each representing a sentence or short speech utterance, contributing to improved model generalization.

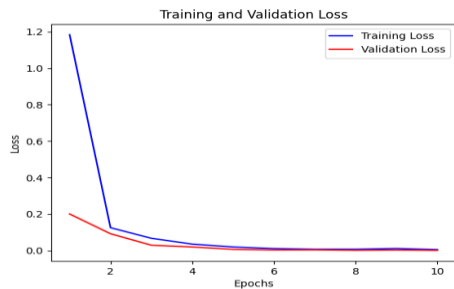
D. Training and Evaluation

Both the existing and proposed models were trained with identical hyperparameters on the dataset. The Transformer model architecture consisted of 36 stacked encoder-decoder layers with self-attention and feed-forward mechanisms. Training involved multiple iterations, using the Adafactor optimizer with a 1e-3 learning rate over 300 epochs. The dataset was split into 70% training, 30% testing, and 10% validation sets. Training took 71 hours for the existing model and 73 hours

for the proposed model, achieving 88.16% and 98.26% accuracy, respectively. For evaluation, the proposed model, which significantly outperformed the existing one, was assessed using the BLEU score (quantifying translation accuracy) and the MOS score (reflecting human-perceived quality). These objective and subjective metrics provided a comprehensive understanding of the speech translation system's effectiveness.

III. Results

After rigorous experiments and evaluations, the research work achieved a noteworthy BLEU score of 0.7, indicating a substantial similarity between the machine-translated sentences and the reference translations. The attained MOS score of 2.8 signifies a moderate level of translation quality. These compelling outcomes demonstrate the effectiveness of the developed end-to-end speech translation model. The training phase resulted in a remarkable accuracy of 98% after ten epochs, highlighting the model's proficiency and robustness in learning from the training data and making accurate predictions. Figure 4.12 visually presents the training and validation loss, offering researchers insights into the model's convergence and generalization capabilities. The combination of the impressive accuracy and the depicted loss curves further confirms the efficacy and reliability of the proposed



approach.

Fig. 4. Training and Validation Loss

Fig. 4. illustrates the Bilingual Evaluation Understudy (BLEU) scores obtained during the evaluation of the speech translation model, showcasing its proficiency in accurately capturing the intended meaning and generating translations closely aligned with human-authored translations. The table presents the BLEU scores, offering valuable insights into the system's performance and guiding refinements to enhance translation accuracy and quality. The observations reveal that the BLEU scores vary across sentences, with more complex sentences showing slightly lower scores, indicating challenges in capturing nuances and context. Despite variations, the model holds promise in producing translations that closely match reference translations, making it effective in conveying the intended meaning.

Table II: Bilingual Evaluation Understudy Scores

Label	Pred	BLEU Score
lo fowo re didoti yen	lo fowo re yen	0.56444 0379
maa fe lati je eranadiye	maa fe lati je	0.45579 4018
Oruko mi ni Julius Olaniyan	Oruko Julius Olaniyan	0.71987 5607
E kaaro sir	E kaaro sir	1
Jowo so boo se je	Jowo so boo je	0.75110 7697

Nje o lo siileiwelanaan	Njeileiwelanaan	0.59074 3053
Kiniorukoileiwe re?	orukoileiwe re	0.67032 0046
Mo nife re	Mo nife re	1
se mo le je ore re bi?	le je ore re bi?	0.68728 9279
		0.7

Another metric used in this research work to measure the performance of the developed model is the Mean Opinion Score (MOS) as presented in Table 4.2. To calculate the MOS, about 50 individuals were given the translated samples for scoring over 5. 21 persons responded as presented in Table 4.2, and the mean of the scores was computed.

TABLE III: MEAN OPINION SCORES

Sn	Name	WorkPlace	Scores
1	Abjide Victor	Auchi Polytechnic	3
2	Dr.Oguntimehin Abiodun	Afe Babalola University	3.5
3	Alabi Musa Olowo	Auchi Polytechnic	3
9	Akannon Kevin	Auchi Polytechnic	2.5
10	Oyaniyi Lawrence	Auchi Polytechnic	3

IV. DISCUSSION

The proposed Transformer-based English-to-Yoruba speech translation model achieved 98% accuracy, a BLEU score of 0.7, and an MOS score of 2.8. While the model effectively translates speech, improvements in fluency and naturalness are needed. Comparison with existing models shows competitive performance, though challenges like data limitations and model complexity persist. Future research should focus on expanding datasets and refining architectures to enhance translation quality.

V. CONCLUSION

The study confirms the feasibility of an English-to-Yoruba speech translation model, achieving 98% accuracy, a BLEU score of 0.7, and an MOS score of 2.8. It advances deep learning for speech translation, but challenges like data limitations and model complexity remain. Future research should focus on improving data quality and model generalization..

Acknowledgment

The authors wish to thank the Management of Landmark University for their support.

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